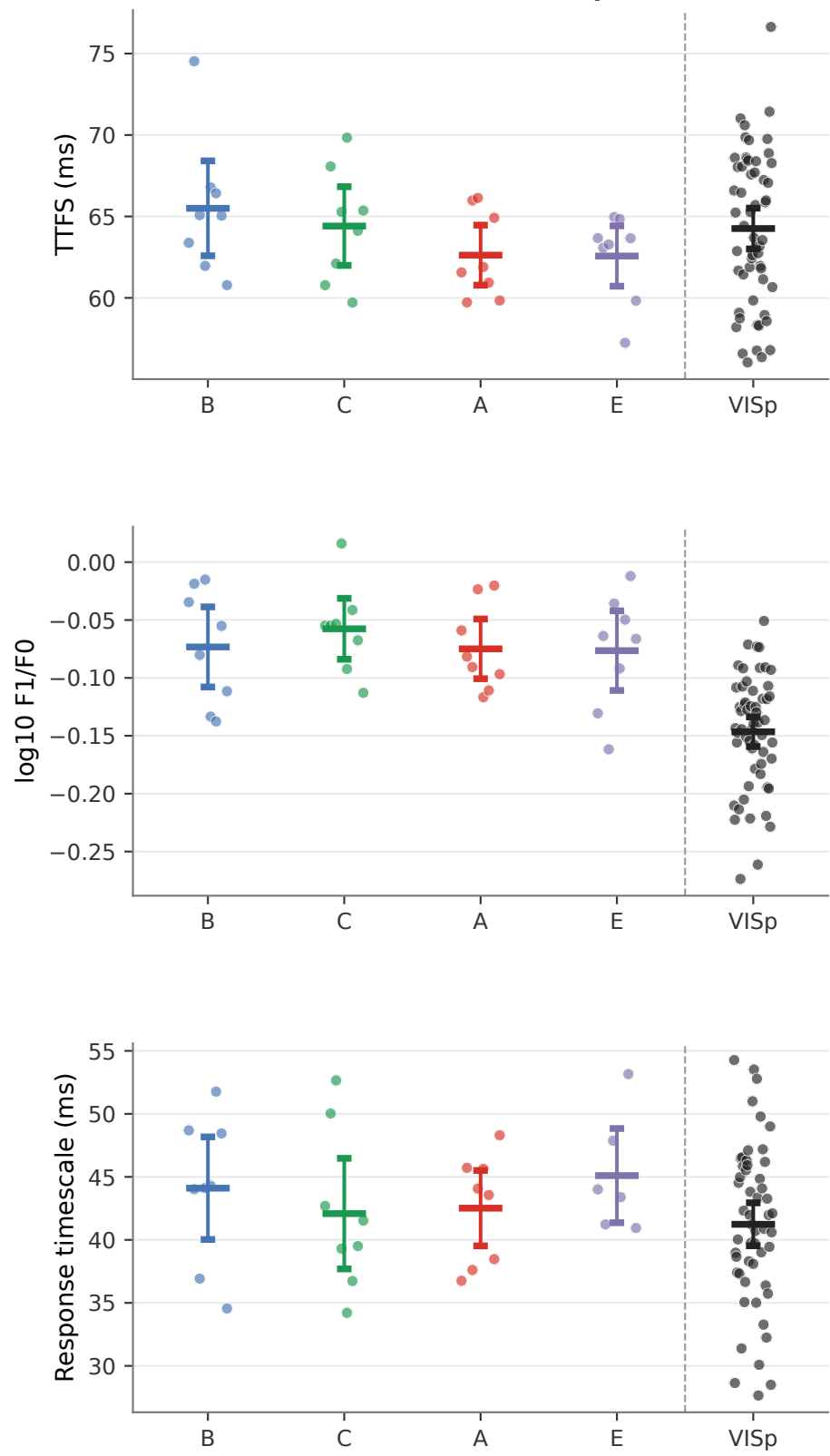


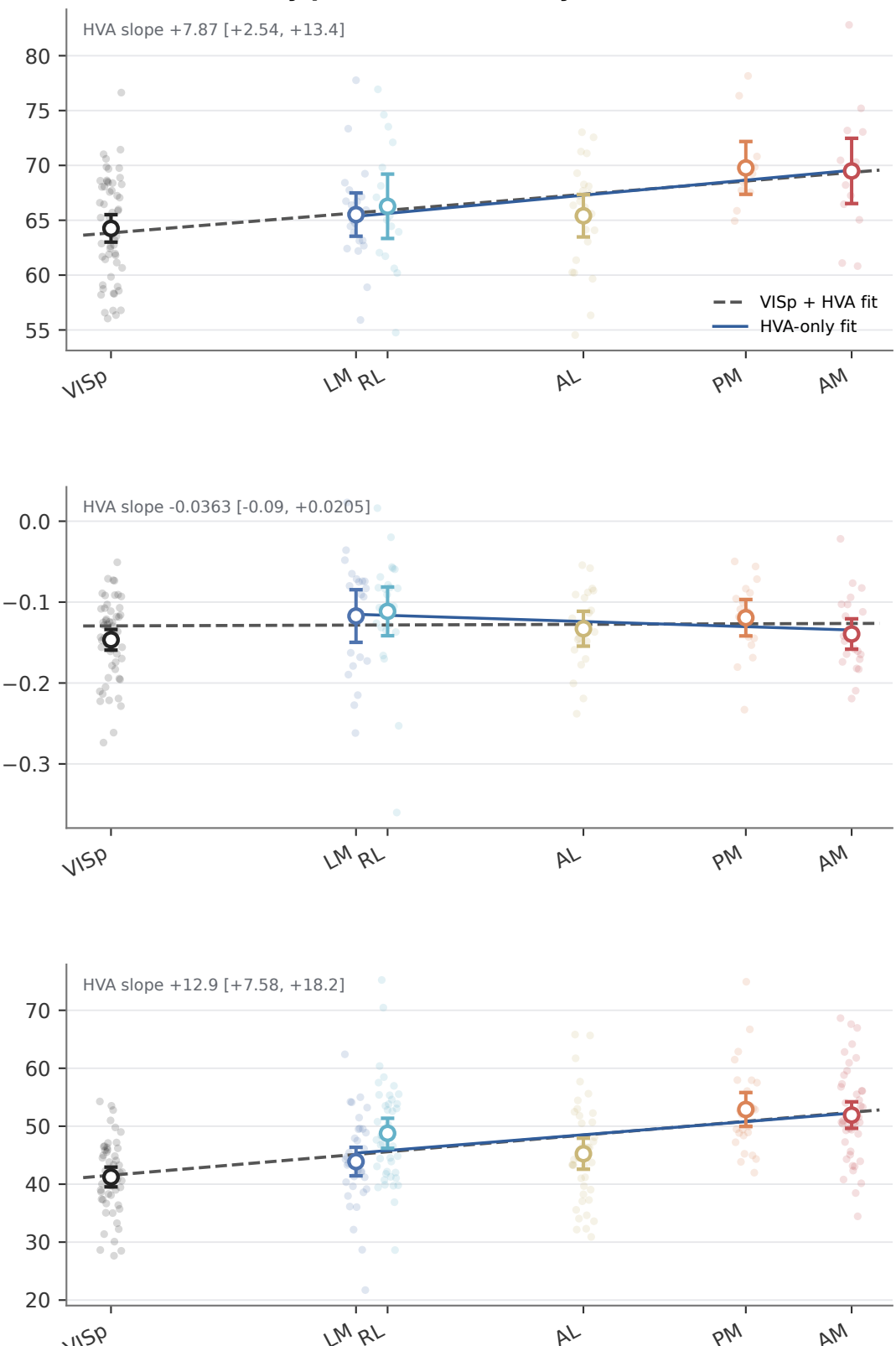
Variant A — hierarchy-led hybrid

Raw session means and hierarchy-score profiles lead; the direct HVA–V1 test remains visible.

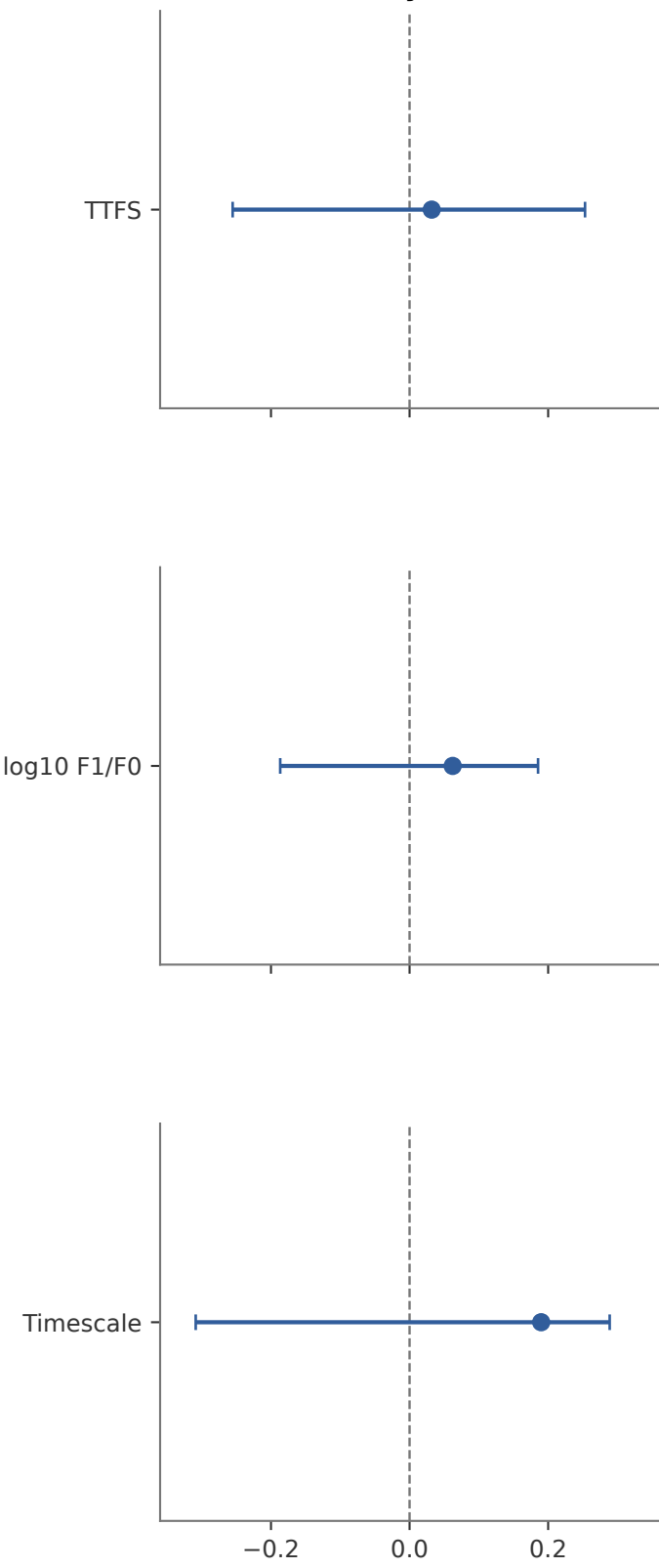
A V1 locations and external VISp reference



B Visual areas by published hierarchy score



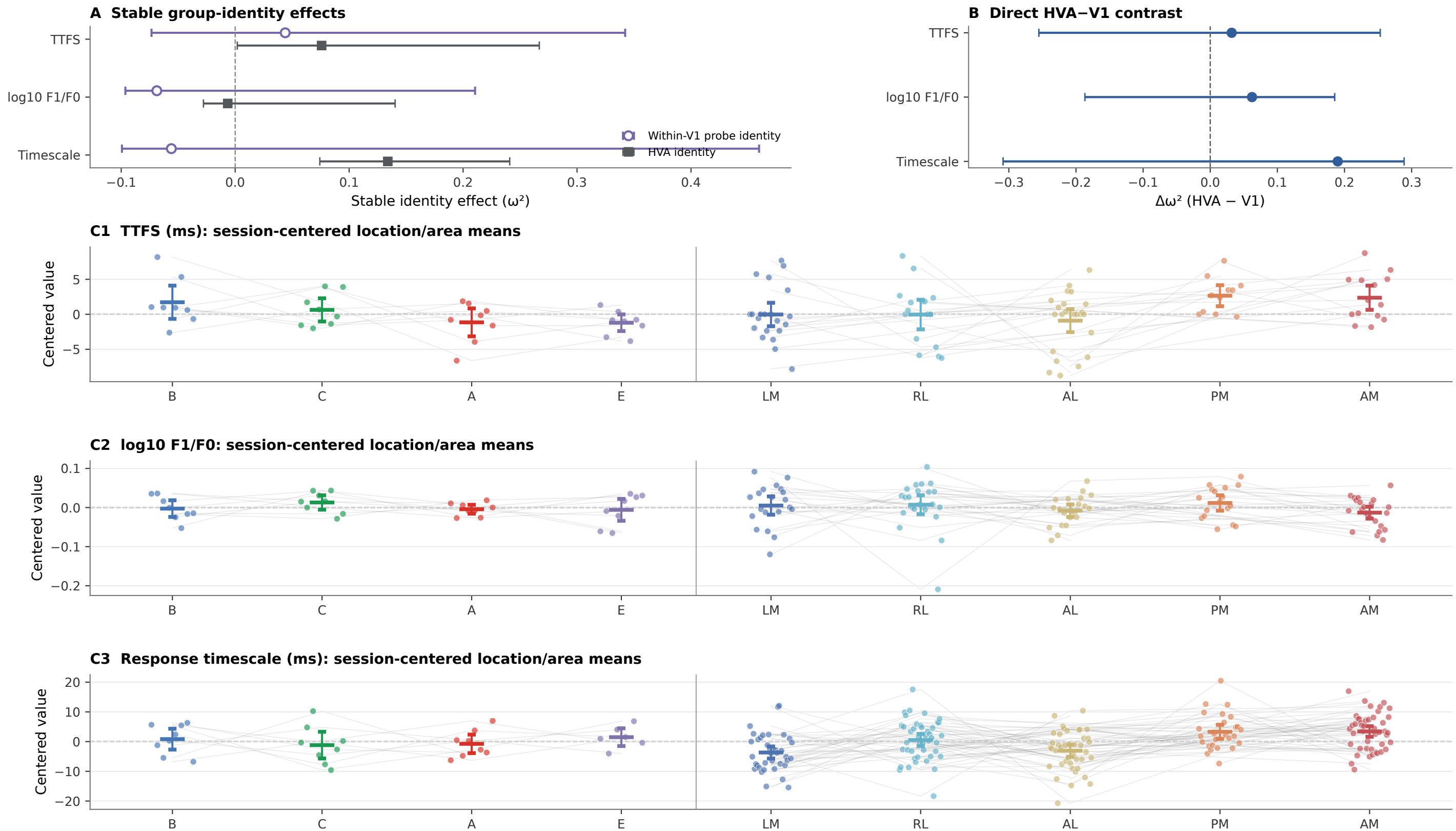
C Direct identity contrast



Session-level estimates; five cortical HVAs (LM, RL, AL, PM, AM); LP excluded. Identity-effect intervals use whole-session bootstrap. V1 probes are categorical locations, not hierarchy scores.

Variant B — benchmark-led

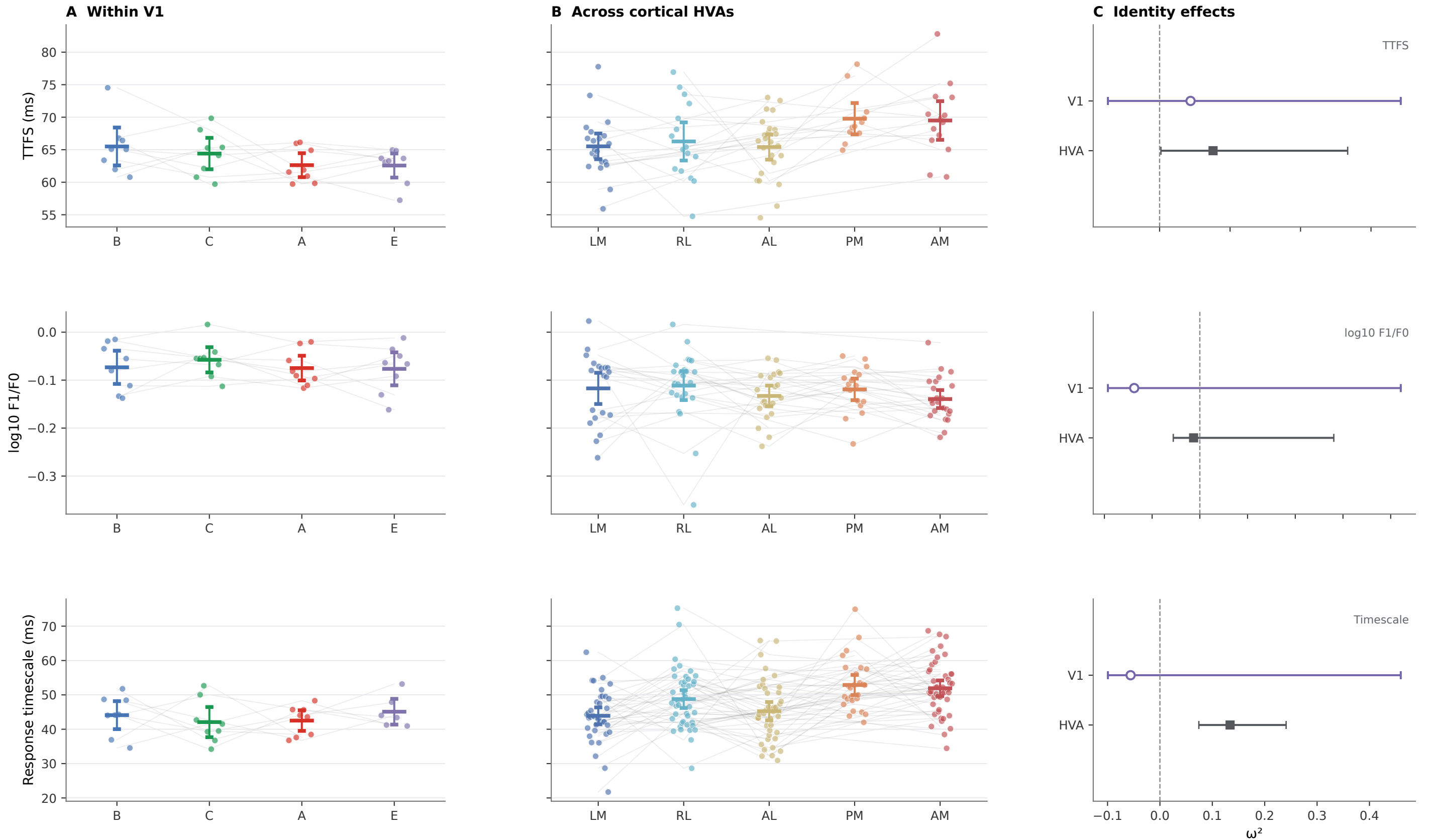
The inferential comparison leads; session-centered data show the observations behind it.



Session-level estimates; five cortical HVAs (LM, RL, AL, PM, AM); LP excluded. Identity-effect intervals use whole-session bootstrap. V1 probes are categorical locations, not hierarchy scores.

Variant C — matched raw-data comparison

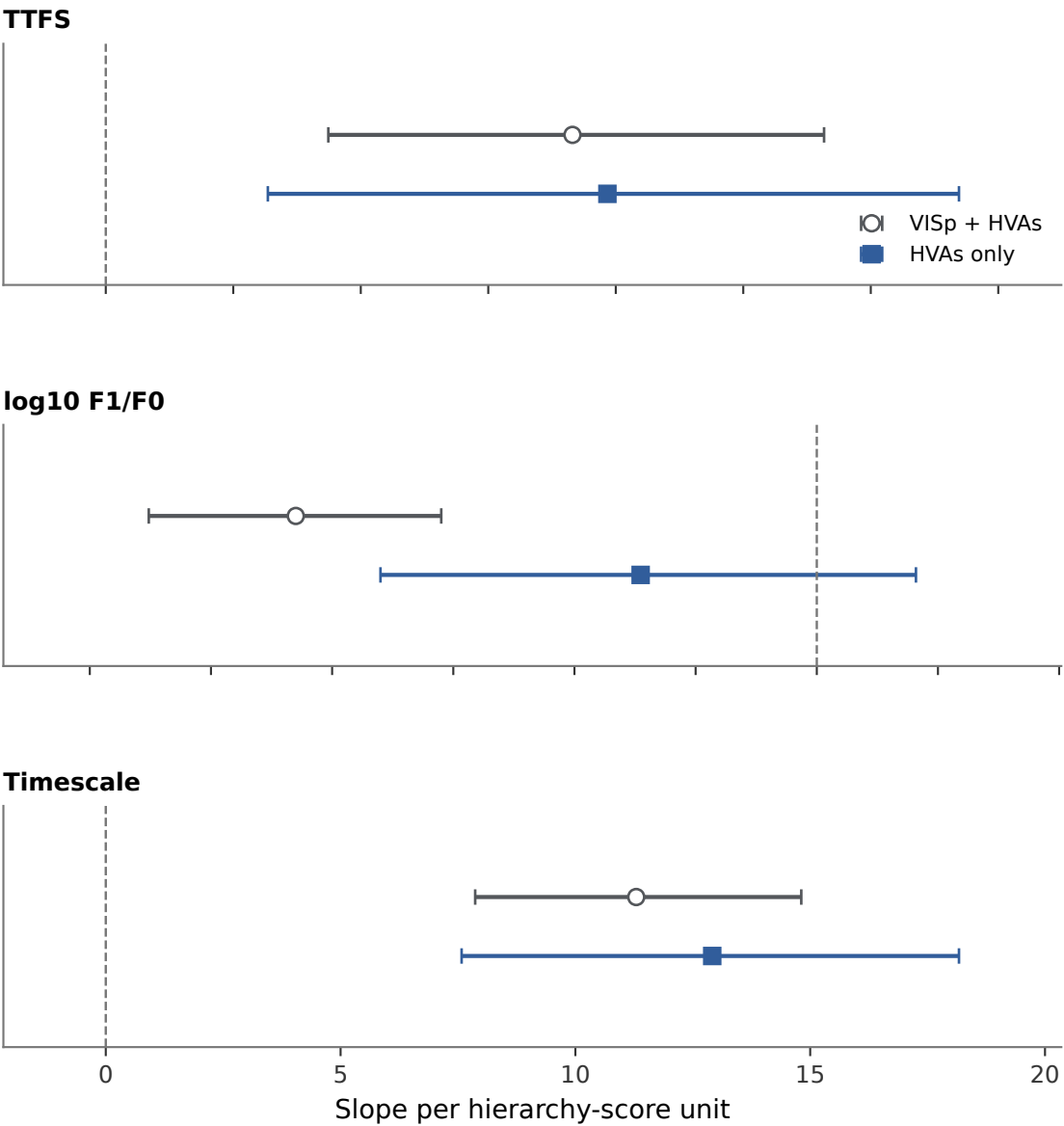
Within-V1 and across-HVA session means share row-wise scales; identity effects provide the summary.



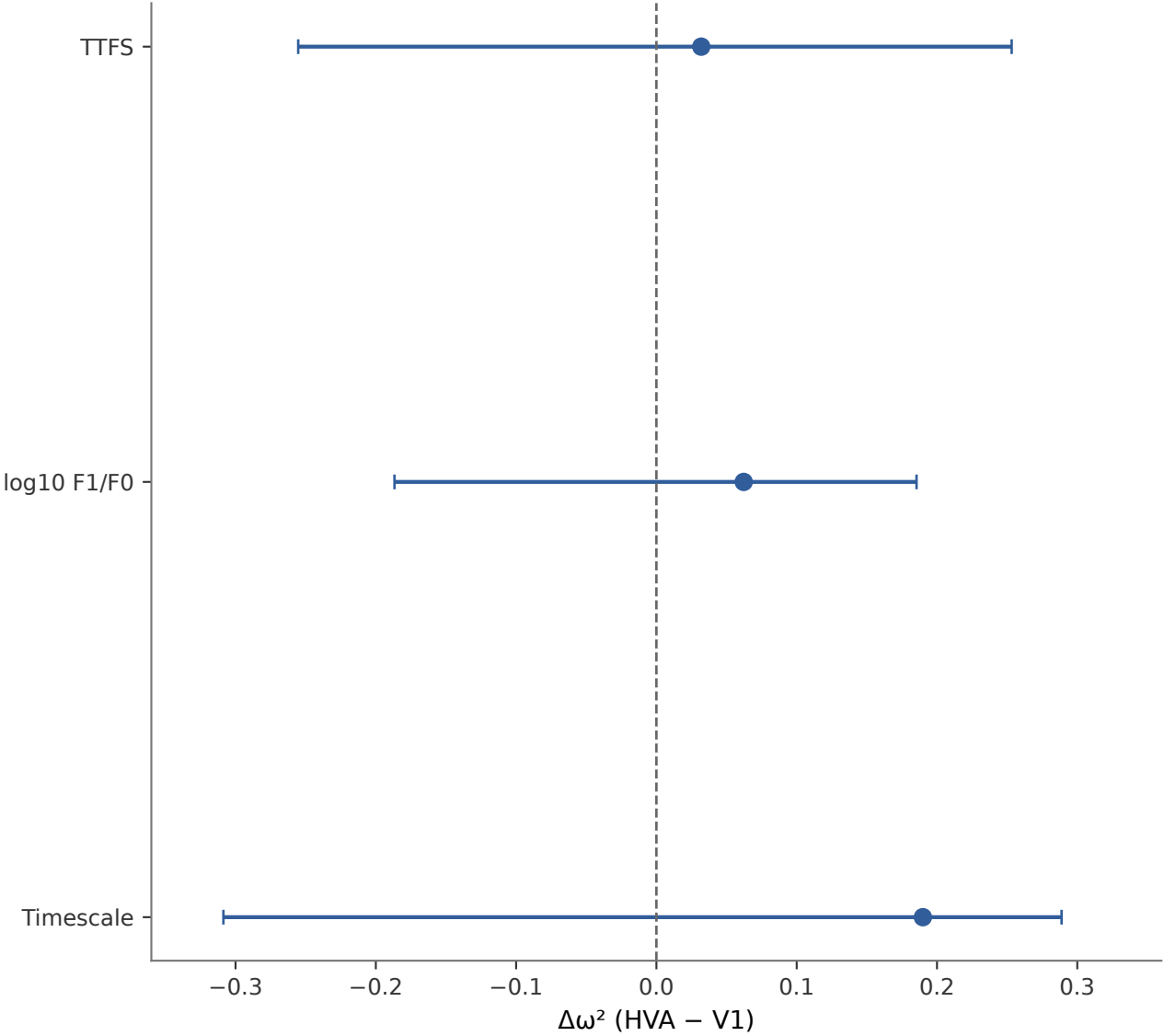
Variant D — compact manuscript summary

Two uncertainty panels for a small main-text footprint; raw session-level panels would move to Extended Data.

A Hierarchy-score associations



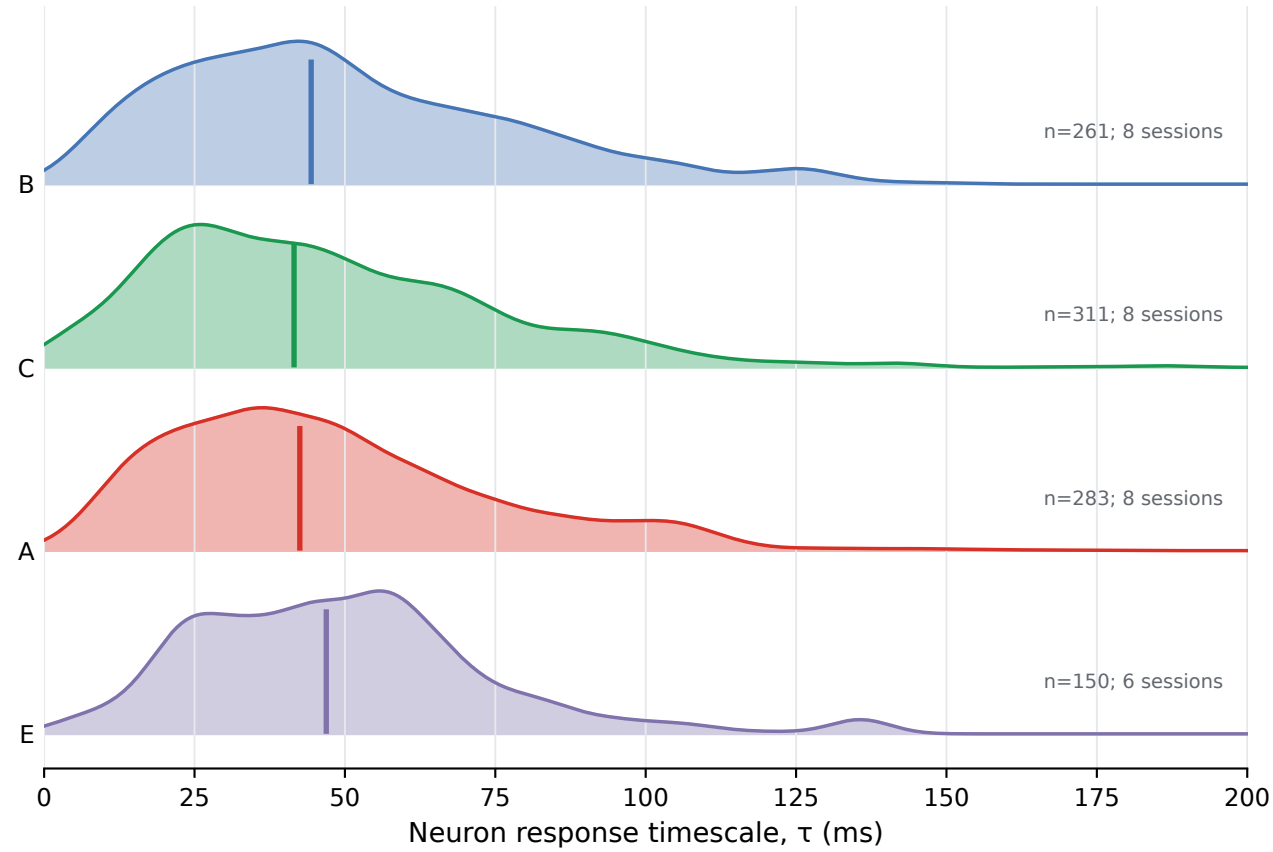
B Does HVA identity exceed within-V1 identity?



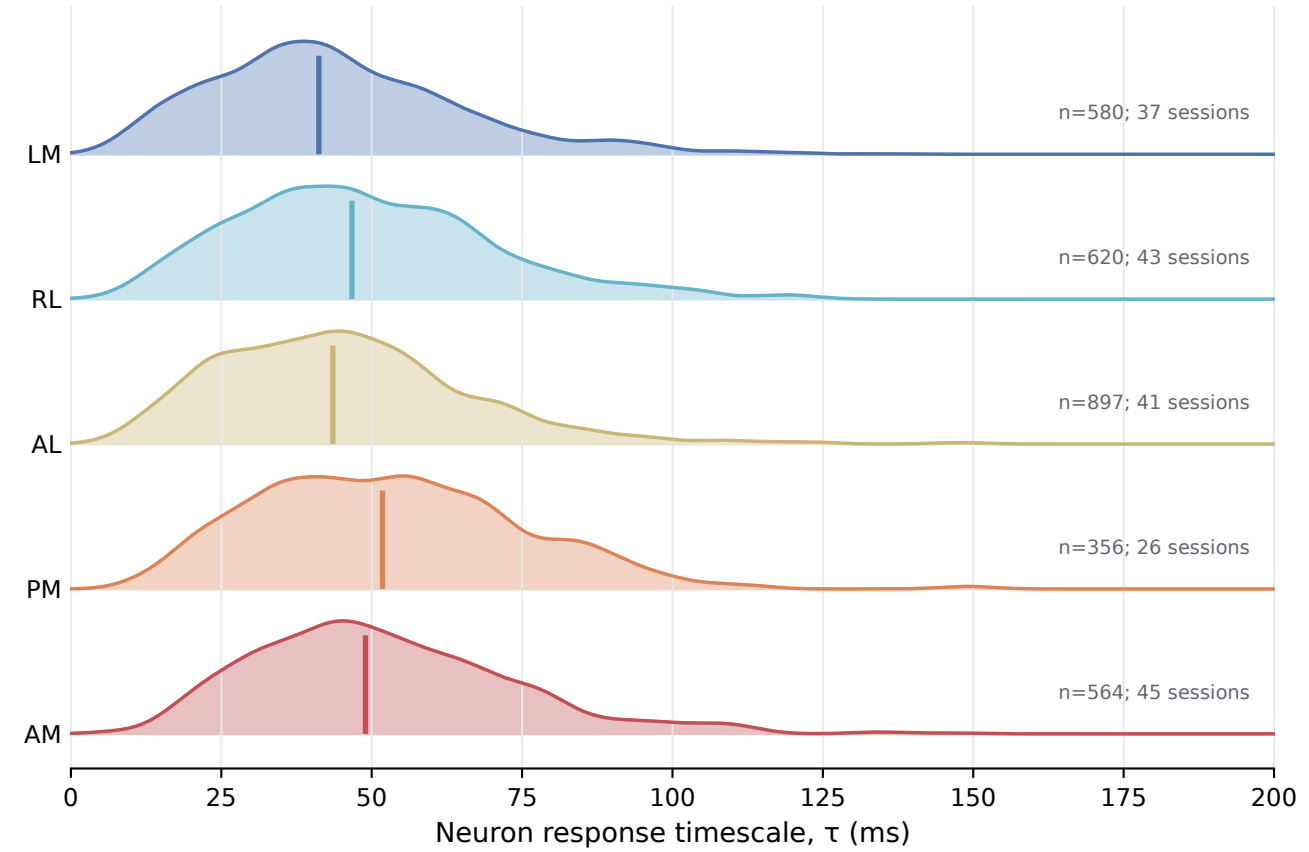
Variant E — matched timescale distributions

Single-neuron ridgelines retain equal session/neuron weight; lower panels show session-level replication.

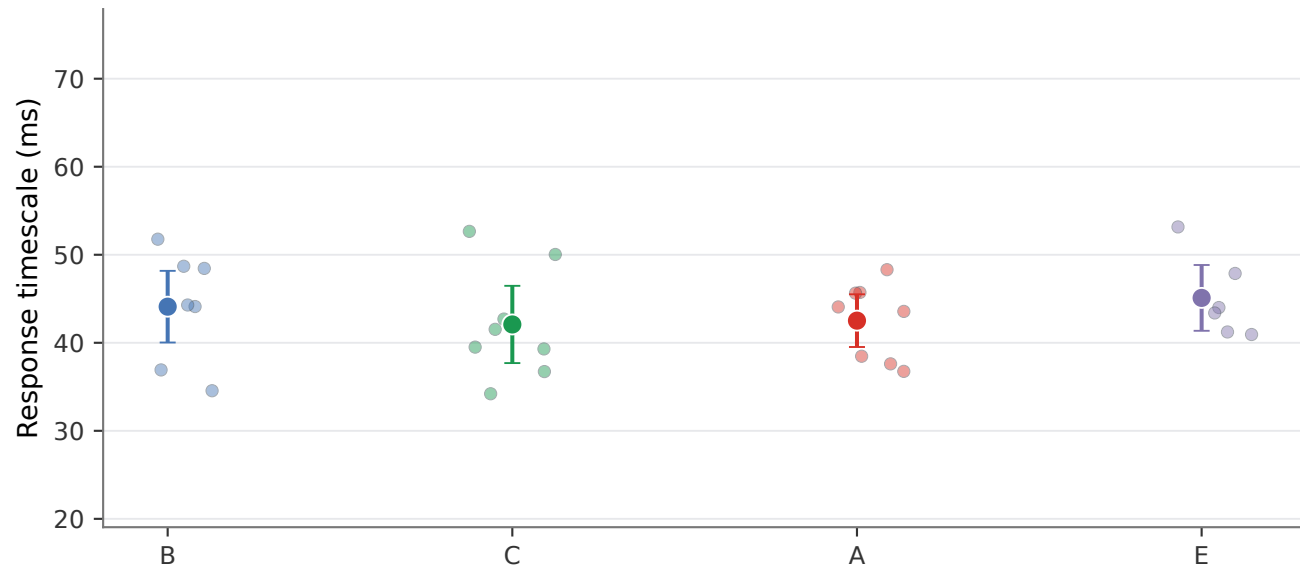
A Full neuronal distributions within V1



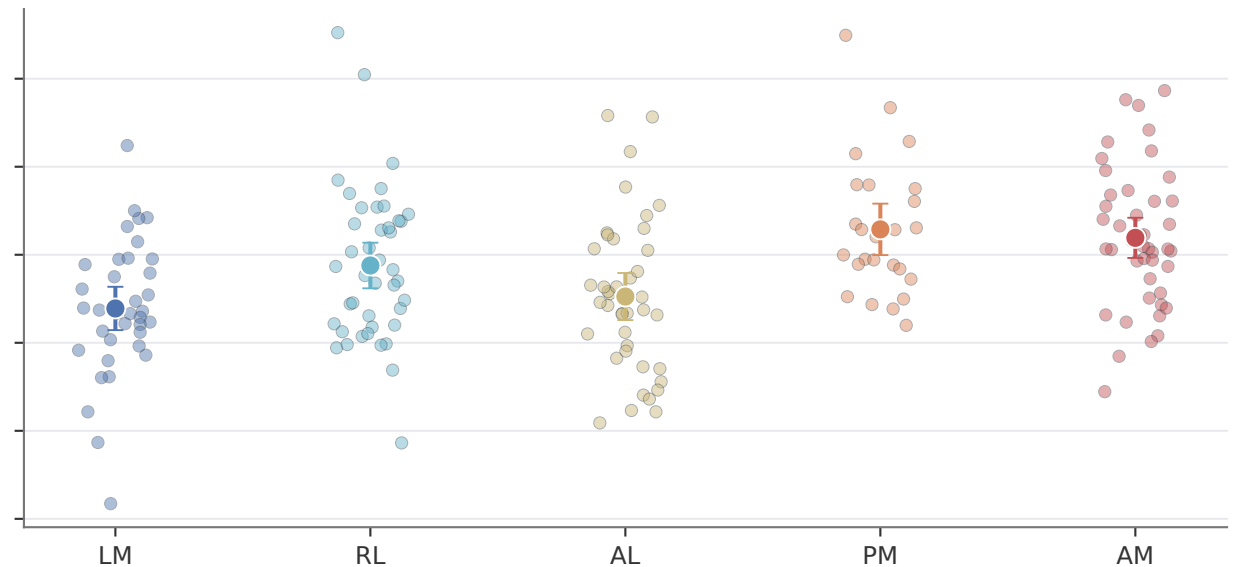
B Full neuronal distributions across HVAs



C V1 session × probe means



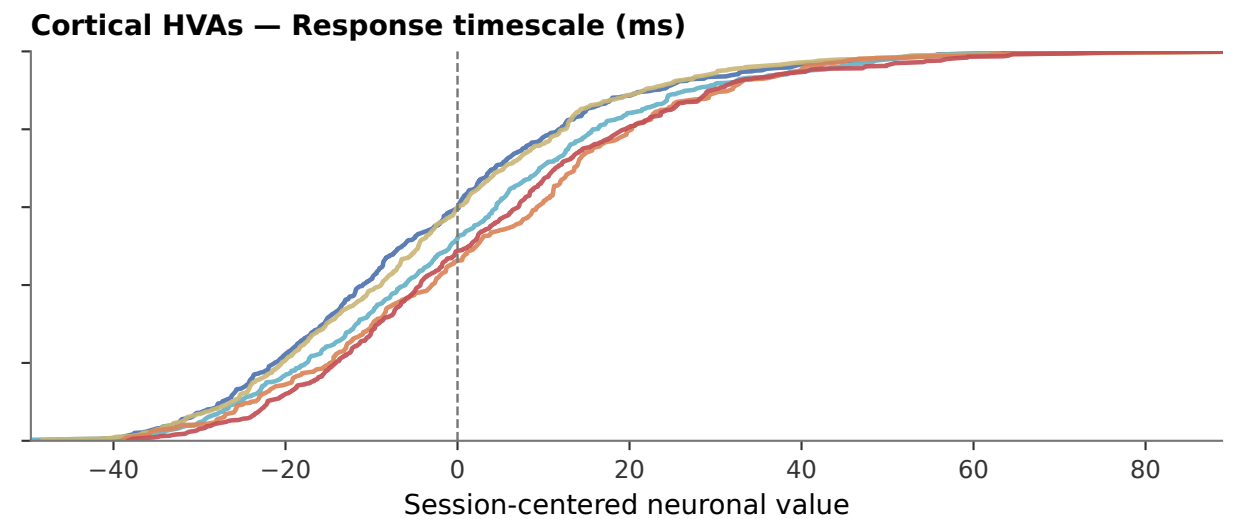
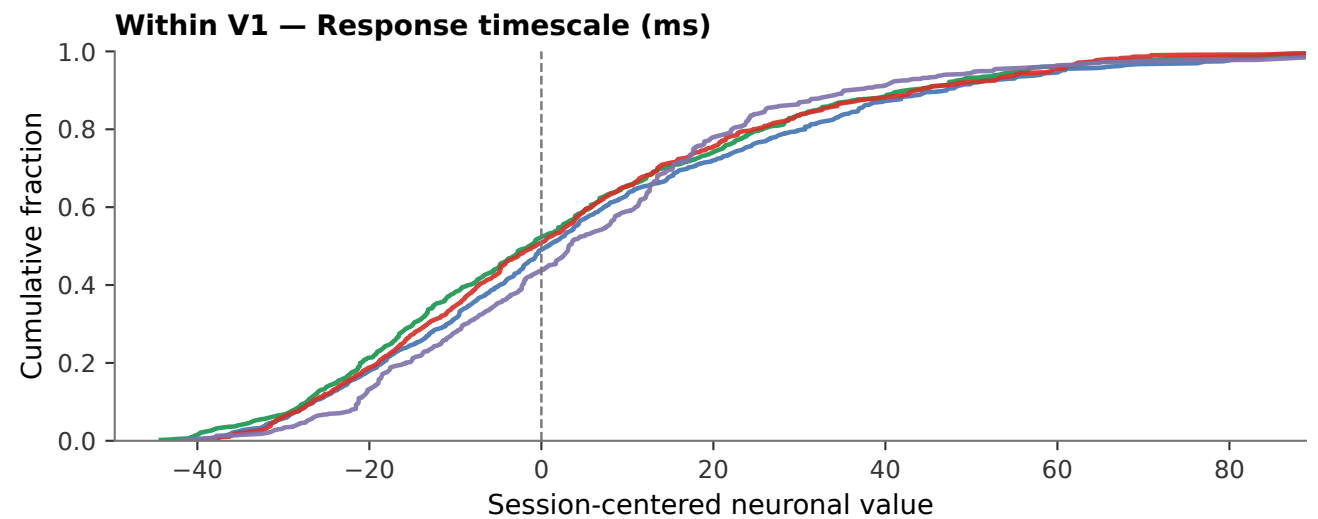
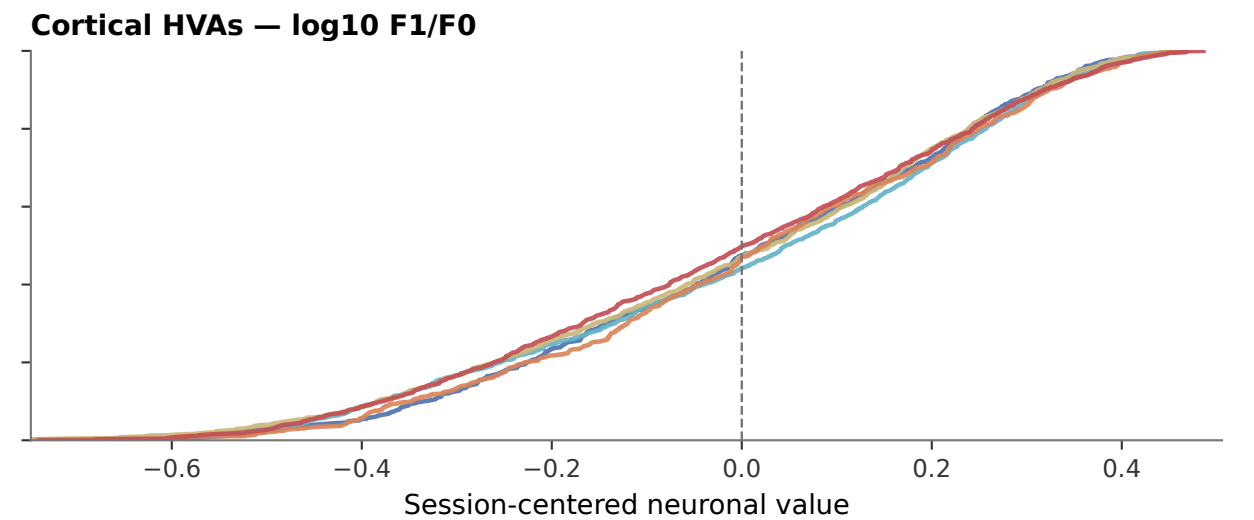
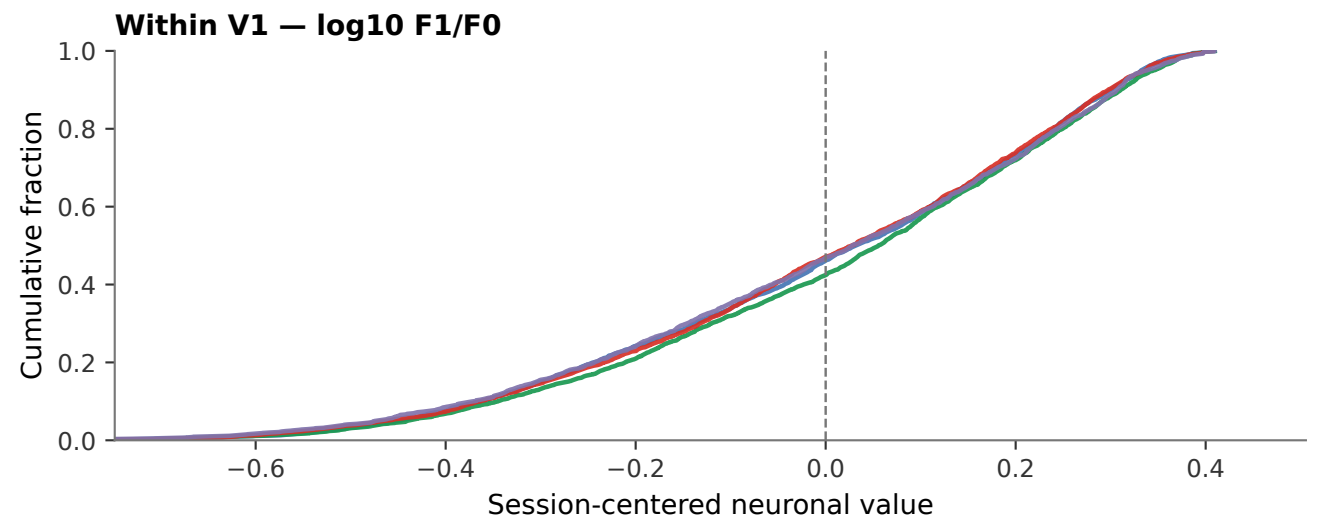
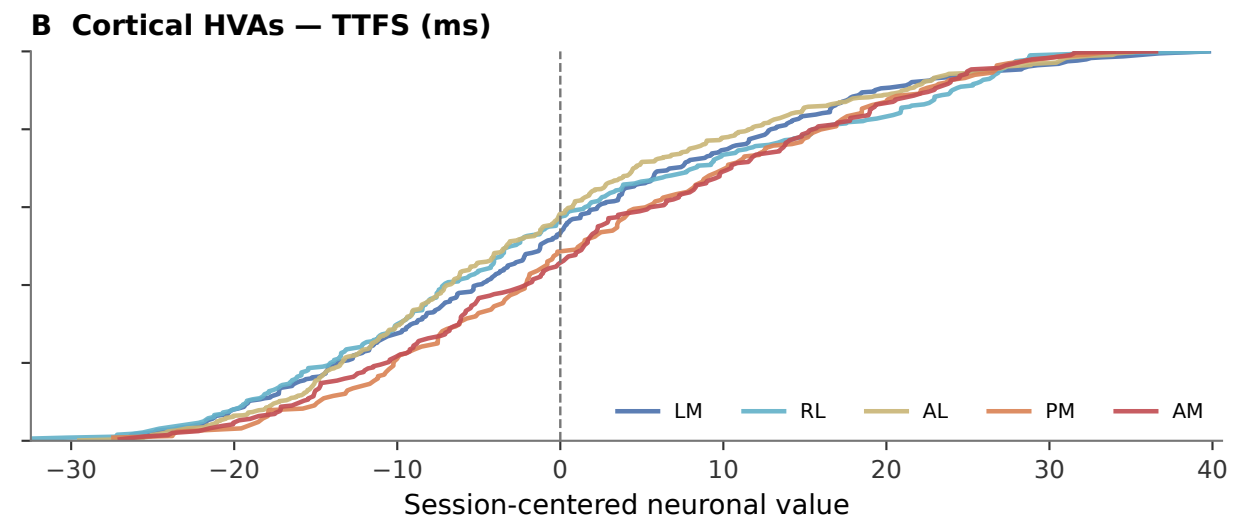
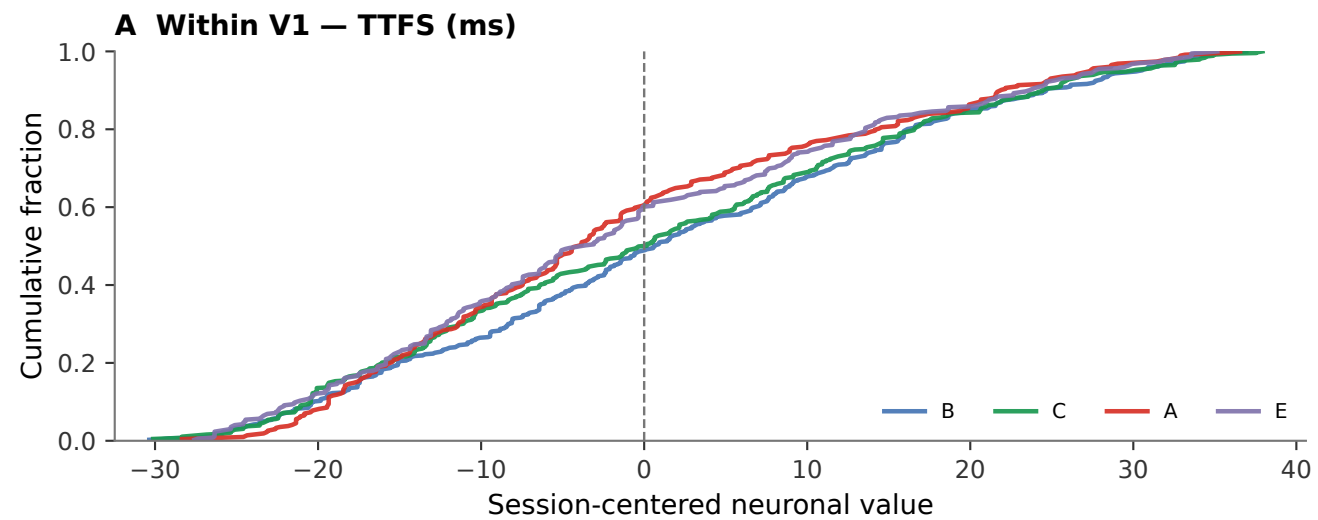
D HVA session × area means



Session-level estimates; five cortical HVAs (LM, RL, AL, PM, AM); LP excluded. Identity-effect intervals use whole-session bootstrap. V1 probes are categorical locations, not hierarchy scores.

Variant F — three-metric full-cell ECDFs

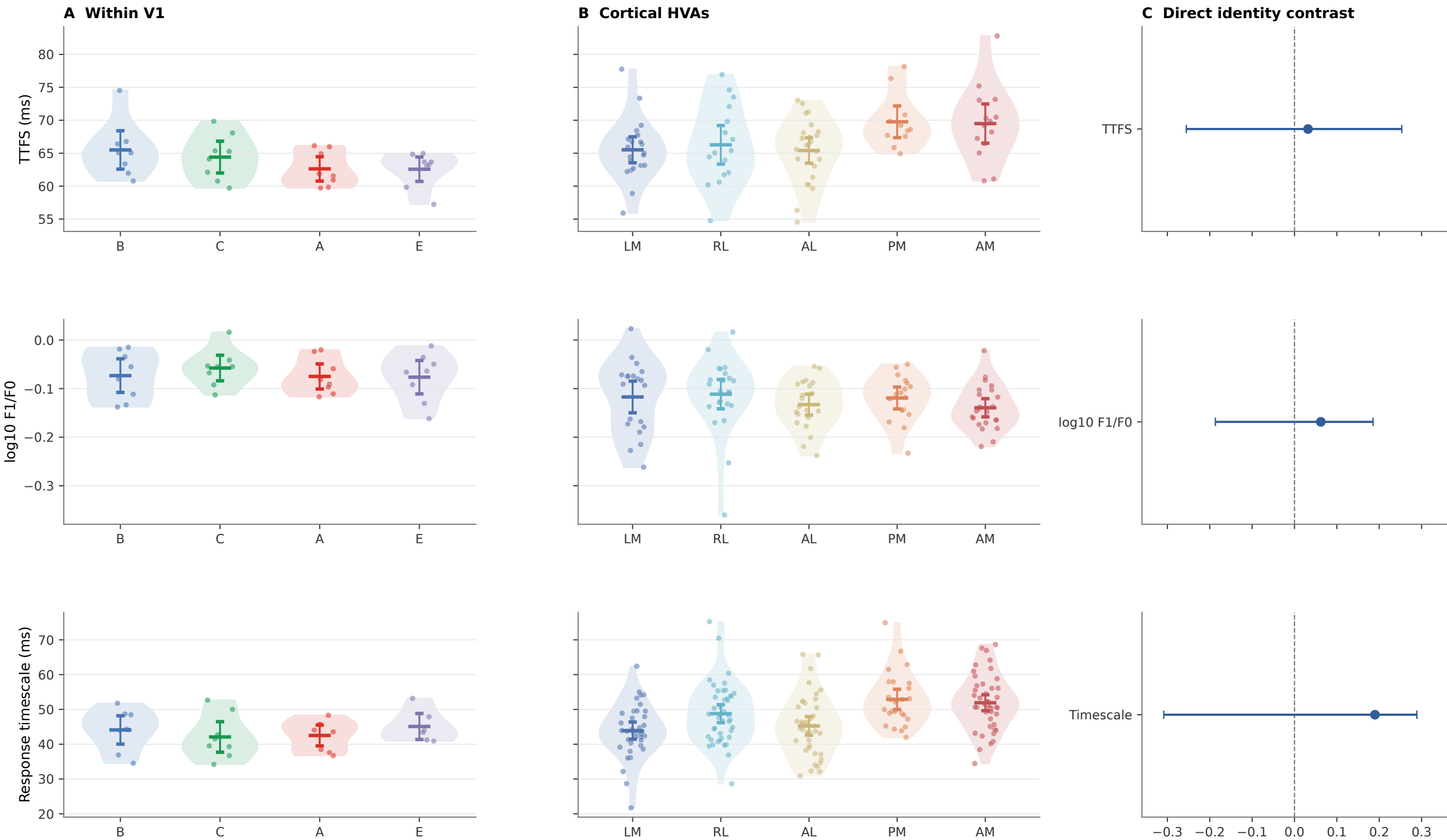
Cells are centered by the frozen session mean and weighted so every session and neuron contributes equally.



Session-level estimates; five cortical HVAs (LM, RL, AL, PM, AM); LP excluded. Identity-effect intervals use whole-session bootstrap. V1 probes are categorical locations, not hierarchy scores.

Variant G — session-level violin comparison

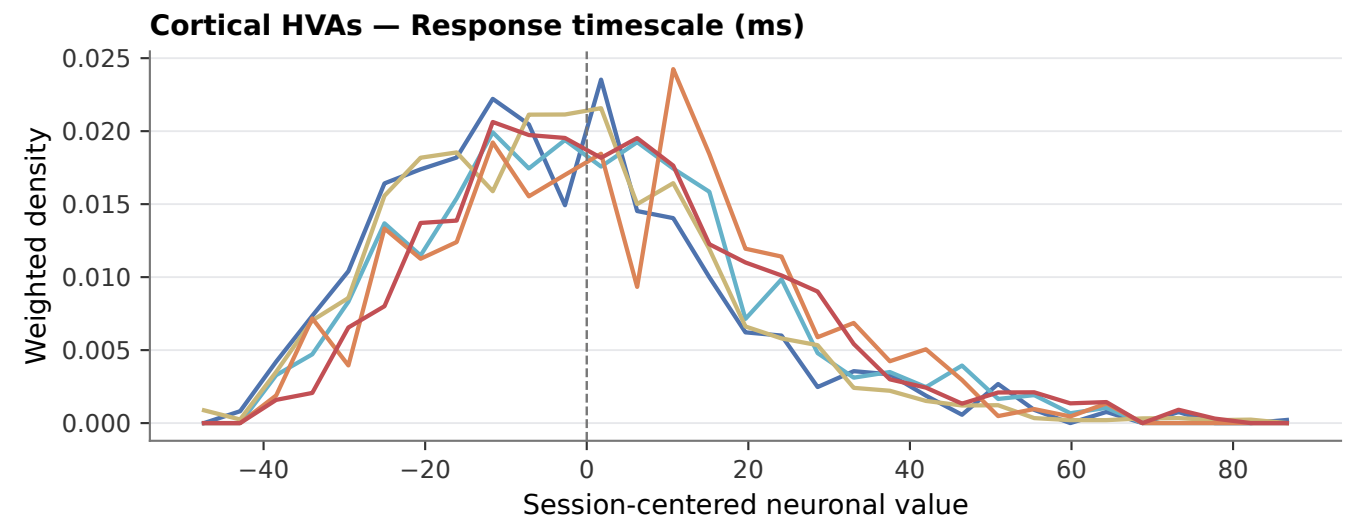
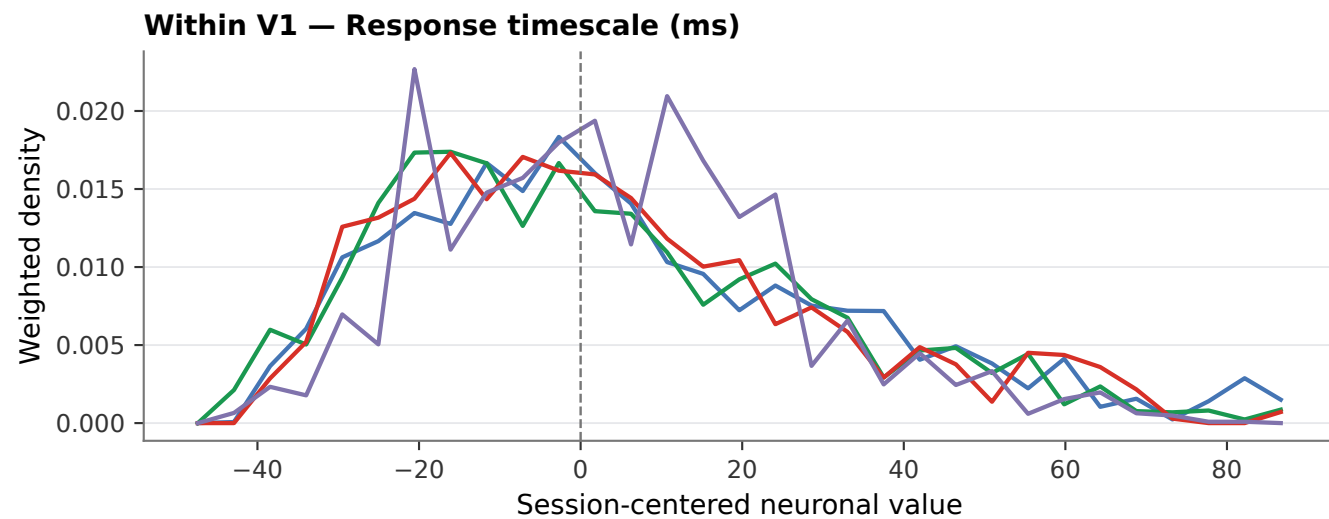
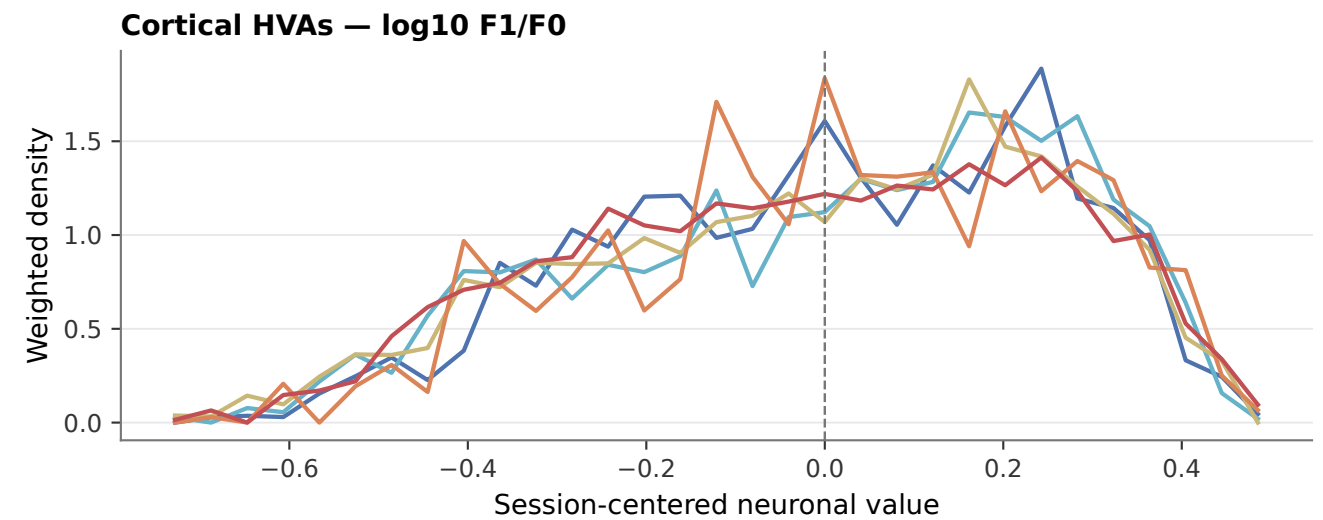
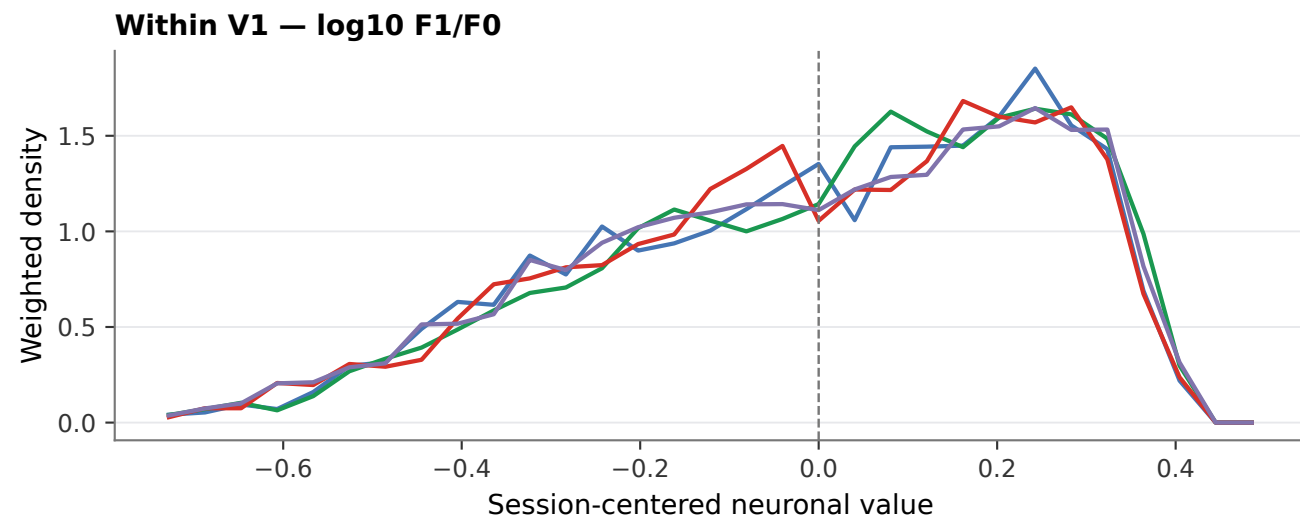
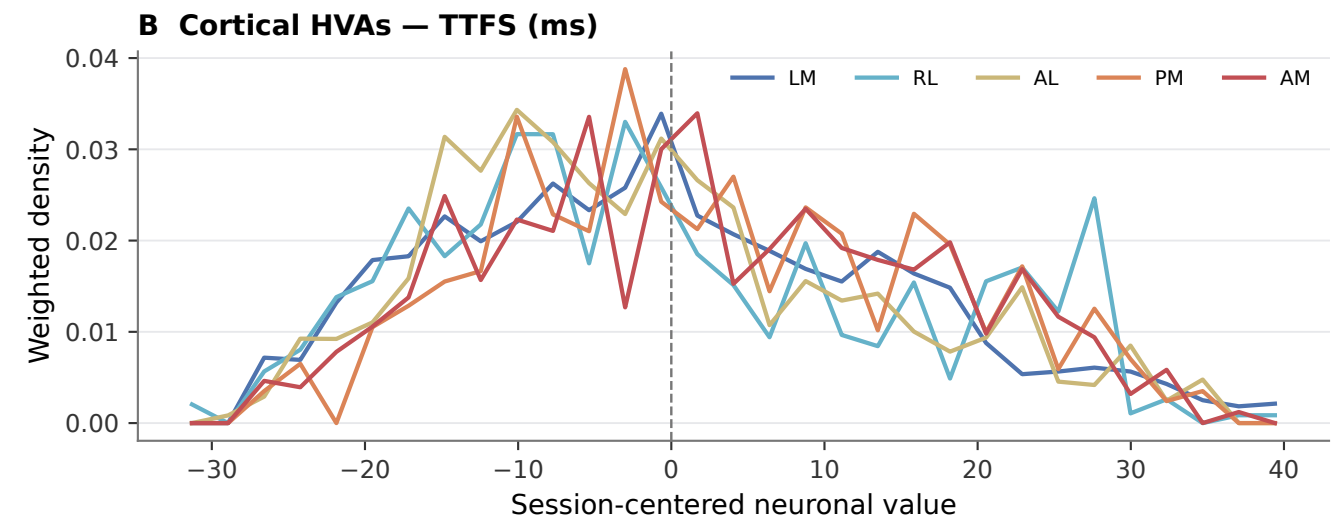
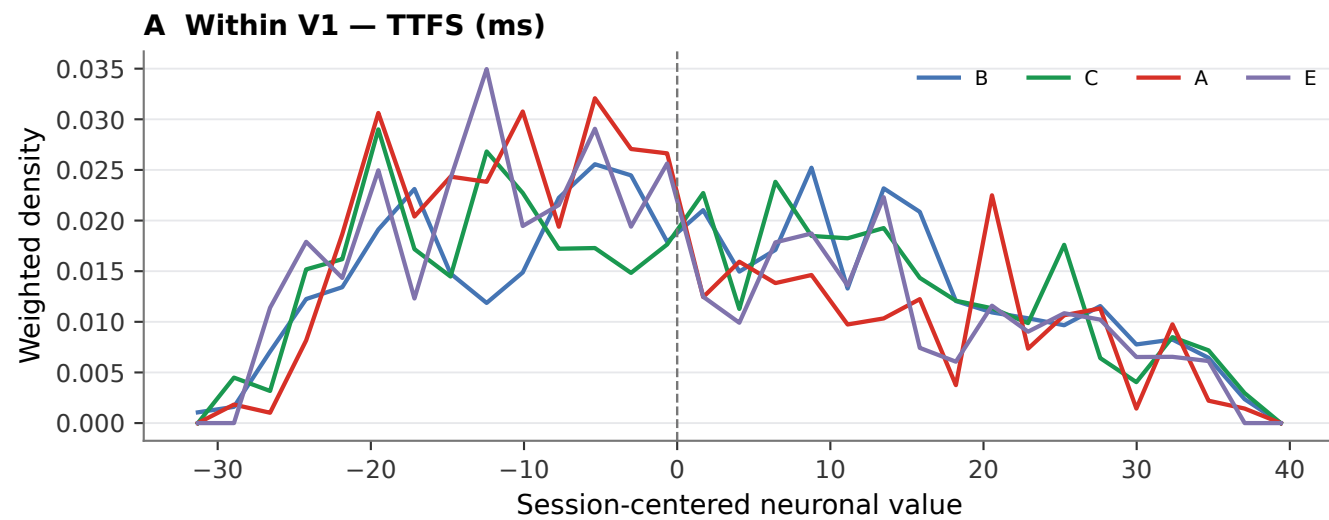
Violin envelopes are descriptive; dots are the inferential session × group observations.



Session-level estimates; five cortical HVAs (LM, RL, AL, PM, AM); LP excluded. Identity-effect intervals use whole-session bootstrap. V1 probes are categorical locations, not hierarchy scores.

Variant H — single-cell histogram small multiples

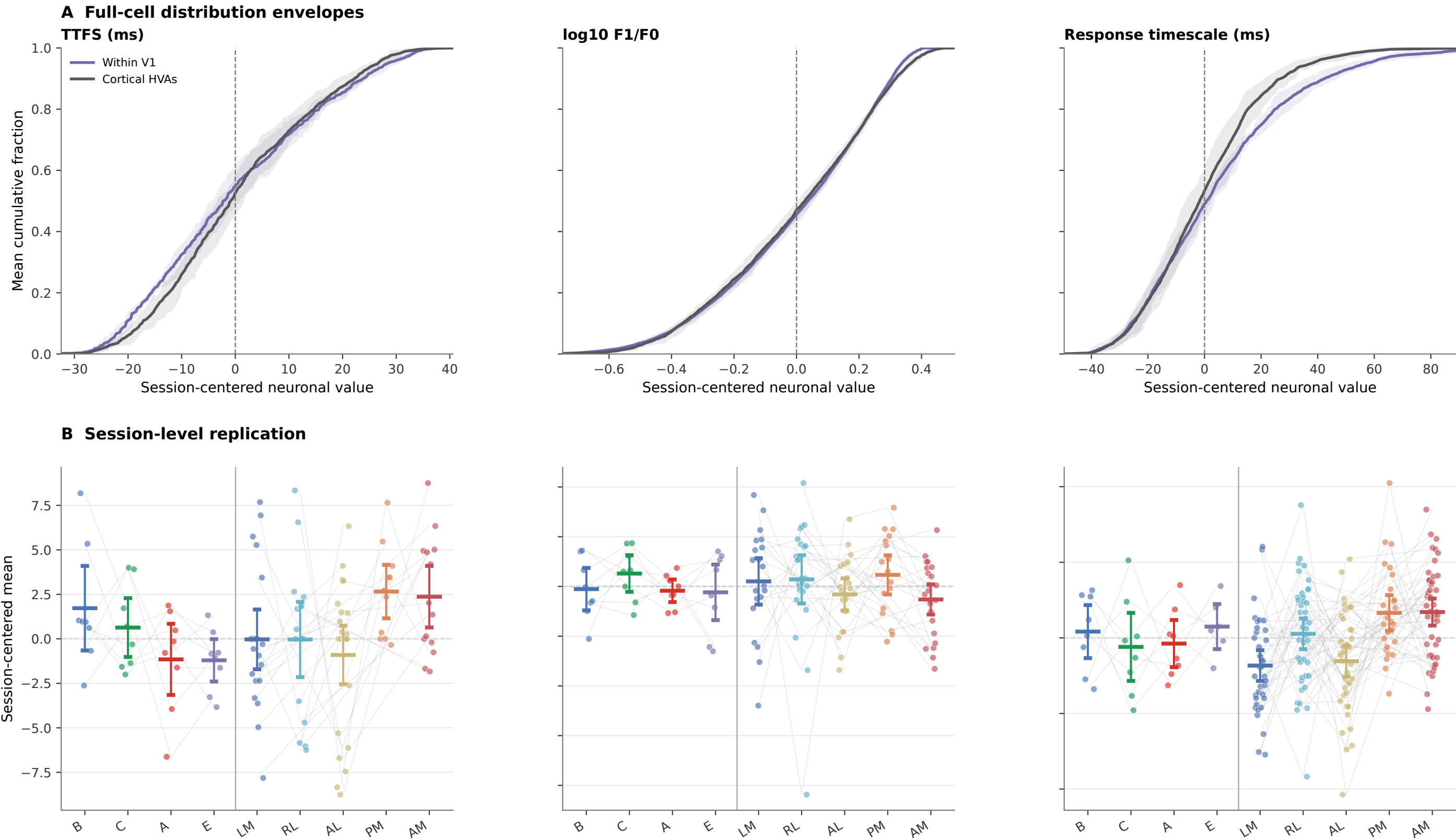
Explicit bins show distribution shape; weights divide repeated draws and equalize neurons and sessions.



Session-level estimates; five cortical HVAs (LM, RL, AL, PM, AM); LP excluded. Identity-effect intervals use whole-session bootstrap. V1 probes are categorical locations, not hierarchy scores.

Variant I — compact distribution evidence stack

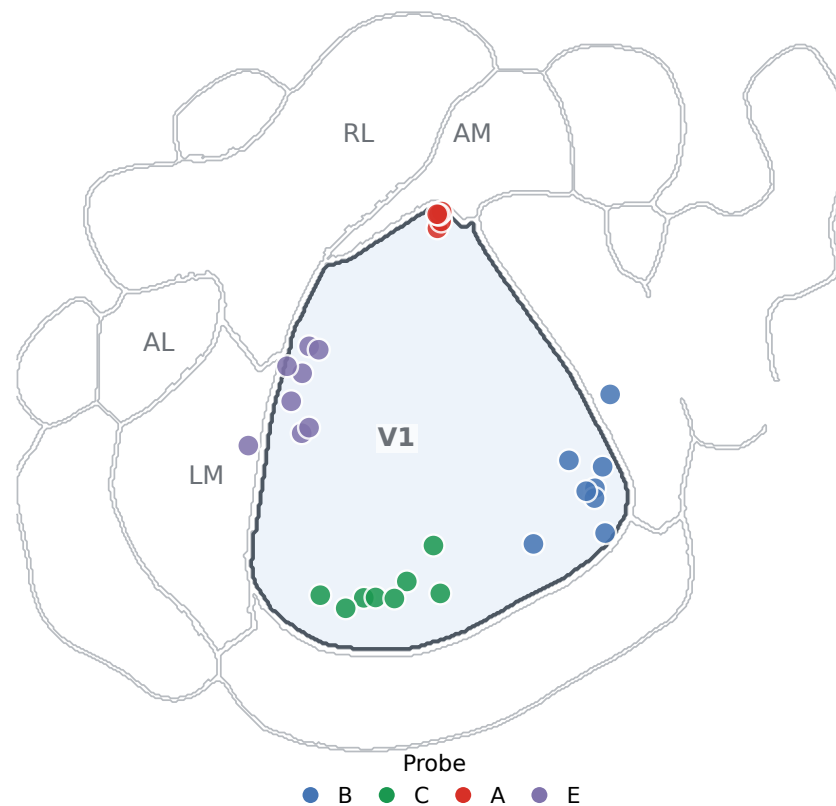
Top: group-balanced neuronal ECDF envelopes. Bottom: every session × location/area mean.



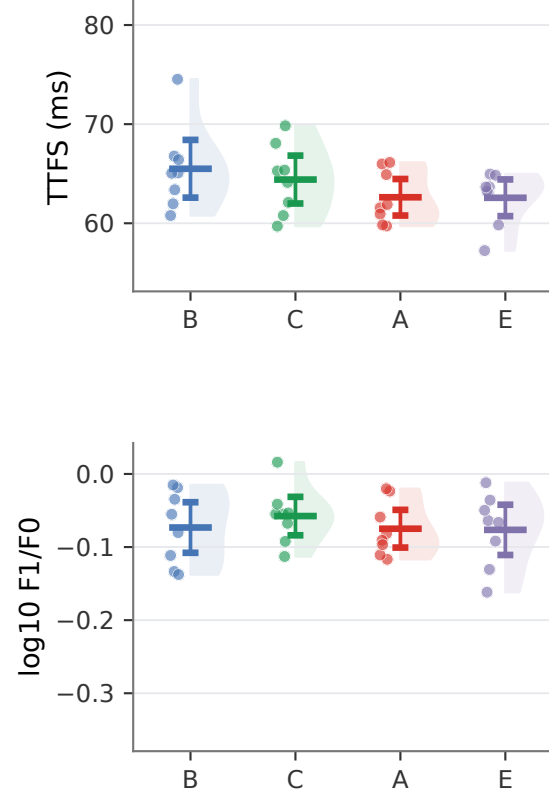
Session-level estimates; five cortical HVAs (LM, RL, AL, PM, AM); LP excluded. Identity-effect intervals use whole-session bootstrap. V1 probes are categorical locations, not hierarchy scores.

A1 MouseV2 V1 penetrations

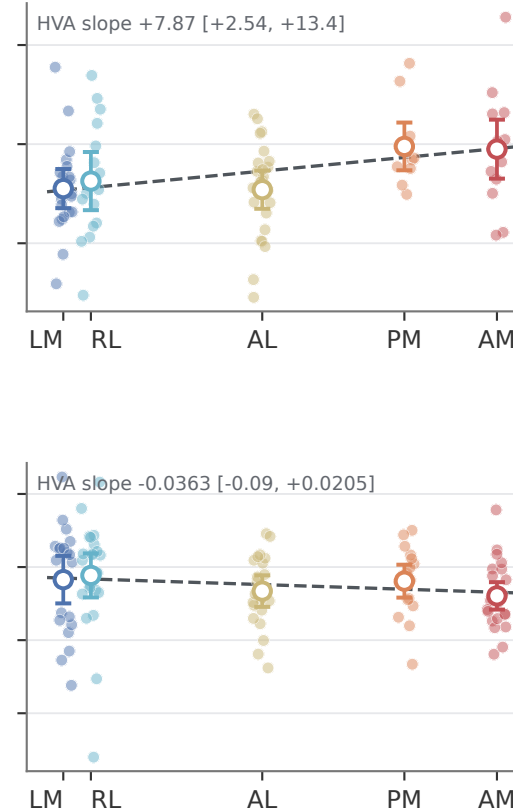
32 registered location estimates · 8 animals · Zhuang common map



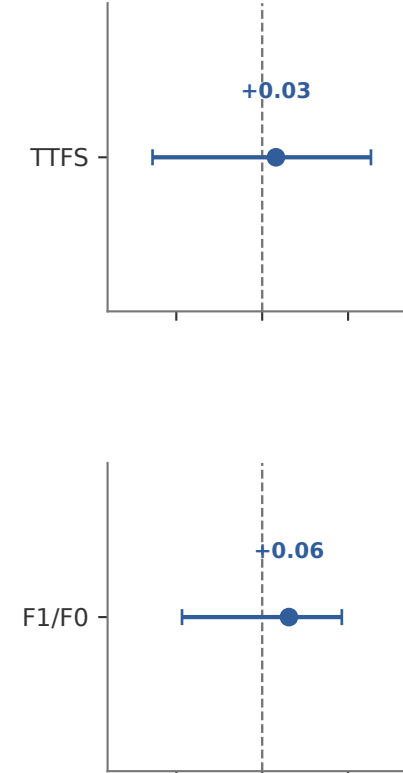
B Within-V1 sessions



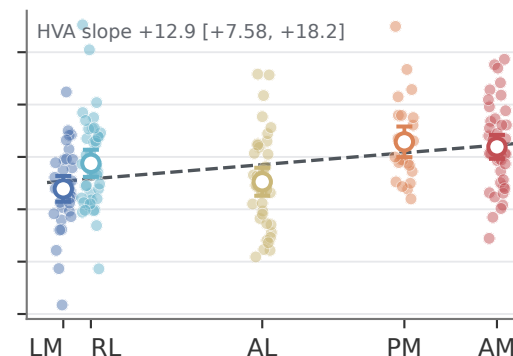
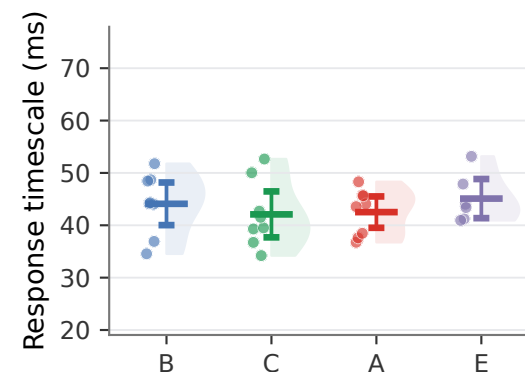
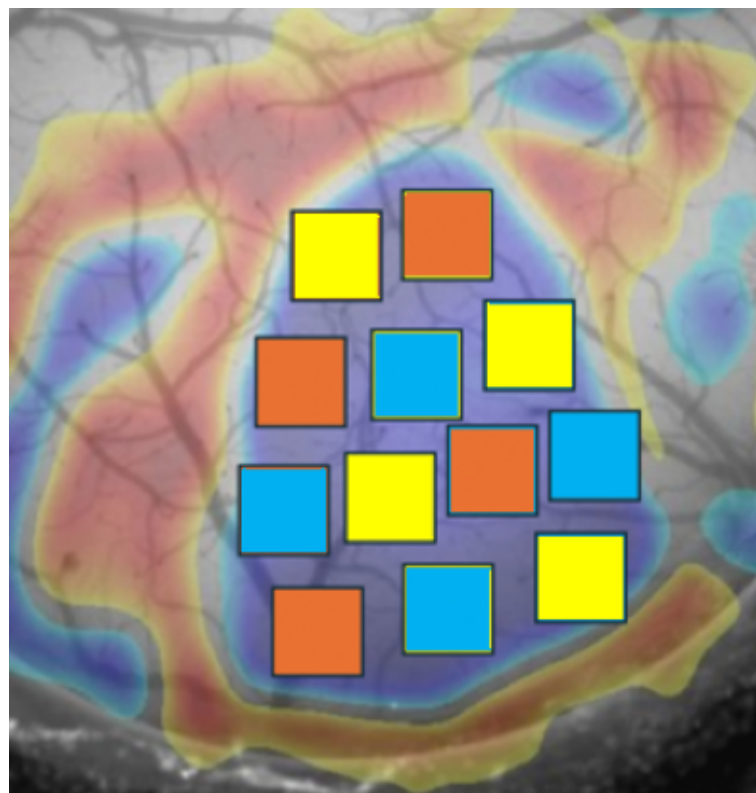
C HVA sessions × hierarchy



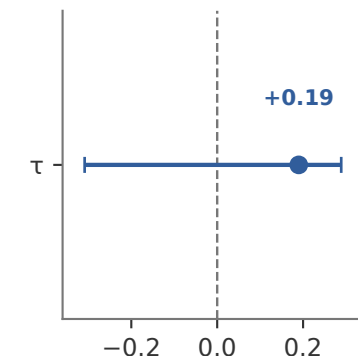
D Direct $\Delta\omega^2$ contrast



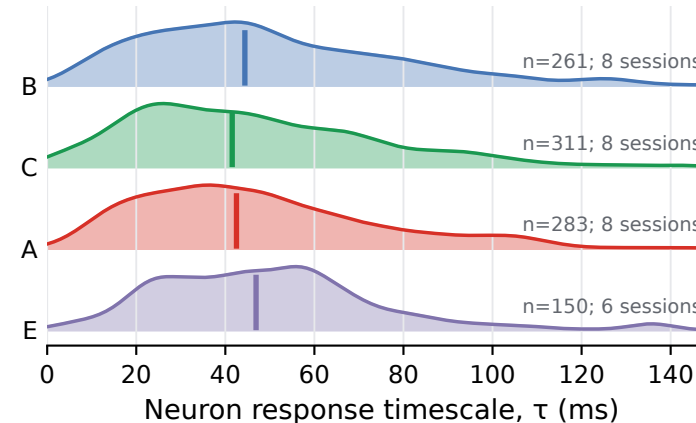
A2 OpenScope imaging schematic



Published inter-area hierarchy score



E1 Within-V1 neuronal timescales



E2 HVA neuronal timescales

